File Permission in Operating System Assignment

The idea is to get yourself familiar with how file permission is being handled in Operating System. You will change file permission for permissionTesting.txt file. You will write down Chmod command to change the file permission based on the questions and show how the permission will look like in terminal.

Chmod Command Sample: chmod ### permissionTesting.txt. (# will be replaced by number).
File Permission in Terminal Sample: -w-rw-rwx filename (-rwx will be replaced based on questions).

Consider the following scenarios. 2*5 = 10 Points

- In a Linux operating system, you have three types of user (Owner, Group, Everyone else). The Owner creates a file name "permissionTesting". This is the permissions situation for "permissionTesting.txt" file at this moment.

  rw- r-- r-- permissionTesting.txt

  (Read = 4, Write = 2, Execute = 1)
  You will use chmod command to change the file permissions.

  Answer the following Questions:

1. What is the file permission for Owner, Group and Everyone else now?

   Owner can read and write but not execute. Group can read but not write or execute. Everyone else can read but not write or execute.

2. Write down a file permission command (chmod) to give Owner Read and Execute permission, the Group can read and write the file and Everyone else has no permission to "permissionTesting.txt" file. After executing the file permission show, how this permission will look like in terminal (As an example, rw-rwx-w- permissionTesting.txt).

   Chmod command: chmod 560 permissionTesting.txt
   File permission in Terminal: r-xrw---- permissionTesting.txt

3. Now modify the file permission again. This time Owner permission can Read and Write the file, Group can write the file and everyone else can read the file. Write down the file permission command (chmod). After executing the file permission show, how this permission will look like in terminal (As an example, rw-rwx-w- permissionTesting.txt).

   Chmod command: chmod 624 permissionTesting.txt
   File Permission in Terminal: rw--w-r-- permissionTesting.txt

4. Now further modify the file permission. The Owner has no permission at all, group can read and execute the file and everyone else can read the file. Write down the file permission command (chmod). After executing the file permission show, how this permission will look like in terminal (As an example, rw-rwx-w- permissionTesting.txt).

   Chmod command: chmod 054 permissionTesting.txt
   File Permission in Terminal: ---r-xr-- permissionTesting.txt

5. Write another file permission command (chmod) to give the owner only to read, write and execute permissions and nobody else has any permission. After executing the file permission show, how this permission will look like in terminal (As an example, rw-rwx-w- permissionTesting.txt).

   Chmod command: chmod 700 permissionTesting.txt
   File Permission in Terminal: rwx------ permissionTesting.txt

   ****Sample and Example: how you should write down the answer. This is not the answer to the questions.****